Separating Dialectal and Register Variation in Usage-Based Construction Grammar

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Construction Grammar (CxG) represents language through form-meaning mappings and traditionally has insisted on *isomorphism*, a one-to-one mapping between the two. Studies of variation have countered this assumption, for example by showing that redundant constructions do not increase processing difficulty (Gardner et al., 2021). Recent work has integrated variation into CxG through the lens of social meaning (Leclercq & Morin, 2023, 2025), raising the question of whether register-based and dialect-based variation differ from a cognitive perspective. This paper hypothesizes that constructional variation is ultimately caused by two very different mechanisms: First, exposure-based variation (dialect) is an artifact of the process by which different populations of speakers learn from different sets of input. Because speakers are aware of who produces which variants, constructions over time take on social meaning. Second, context-based variation (register) results from the conventionalized association between specific constructions and specific registers. This association is arbitrary and we thus expect that registers will vary slightly across populations. Our hypothesis is that the overall profile of each will differ in its sociocognitive salience (Morin, *in press*).

Data. We compile corpora which represent register-specific production from different populations: written Tweets (Dunn, 2020: totalling 2.05 billion words) and spoken YouTube transcripts (Coats, 2023: totalling 436 million words) from five inner-circle dialects of English. These corpora are aligned across 392 local metro areas to capture variation across local communities as well as national groups.

Methods. We leverage recent advances in Computational Construction Grammar (CCG) to systematically study variation in constructions. CCG is an unsupervised paradigm that acquires, represents, and arranges constructions within a grammatical network from corpus data alone (Dunn, 2017). It simulates the usage-based learning of language, starting from surface-level chunks which are then increasingly generalized given distributional information to form more and more abstract representations (Dunn, 2024). Given these constructional features, we take a classification approach to modelling variation: this kind of model finds those constructions which differ in usage by population or register (Dunn, 2018, 2019). The model is validated on a held-out corpus, where higher predictive accuracy indicates better representations of variation. Once the classifier has been validated, we use its feature weights as a proxy for the salience or predictiveness of constructions for each source of variation.

Results. The classification results for validation are shown in Table 1. First, register variation is generally stronger and more pervasive in the grammar than regional variation, as demonstrated by the higher accuracy of classifiers. Second, register variation is spread evenly across the grammar, while regional variation manifests primarily in lower-level concrete units, as shown by the predictive power of different kinds of constructions. Third, additional country-specific classifiers reveal that the two sources of variation have largely independent feature weights, notwithstanding slight regional effects in the association between constructions and registers.

Discussion. To account for these results, we hypothesize that these two sources of variation operate along a continuum of sociocognitive salience, with dialect much less salient than register. This account is important for connecting constructions with linguistic variation.

Table 1. Validation Results by Region in the Grammar. *Each cell represents classifier performance for a specific part of the grammar (columns) for a specific kind of variation (rows). Higher values indicate better prediction accuracy which then represents a better characterization of grammatical variation. The grammar is divided in three ways following Dunn (2024): level of abstraction looks at smaller or larger families of constructions; order of emergence looks at simple vs complex slot-constraints; and degree of centrality distinguishes between core and peripheral constructions. Cross-fold validation is used to estimate the confidence of each type of performance, here represented by minimum and maximum bounds.*

	By Level of Abstraction					
	First-Order (Concrete)		*Third-Order*		*Fourth-Order (Abstract)*	
	Min F1	*Max F1*	*Min F1*	*Max F1*	*Min F1*	*Max F1*
Dialect	0.97	0.97	0.72	0.72	0.48	0.48
Register	1.00	1.00	1.00	1.00	1.00	1.00
	By Order of Emergence					
	Early-Stage		*Mid-Stage*		*Late-Stage*	
	Min F1	*Max F1*	*Min F1*	*Max F1*	*Min F1*	*Max F1*
Dialect	0.90	0.90	0.81	0.81	0.97	0.97
Register	1.00	1.00	1.00	1.00	1.00	1.00
	By Degree of Centrality					
	Peripheral		*Middle*		*Core*	
	Min F1	*Max F1*	*Min F1*	*Max F1*	*Min F1*	*Max F1*
Dialect	0.97	0.97	0.79	0.79	0.59	0.59
Register	1.00	1.00	1.00	1.00	1.00	1.00

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